

Titanic Survival Analysis

Machine Learning Report | Preprocessing · EDA · Modeling · Dashboard

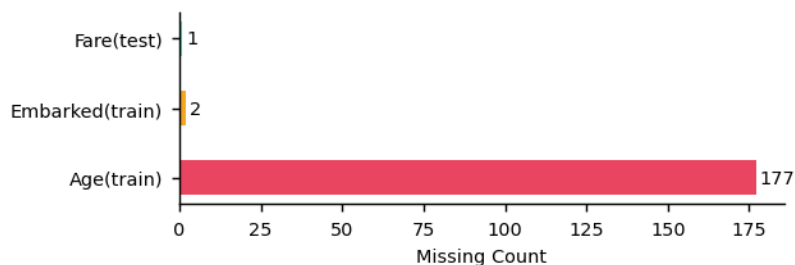
1. Data Preprocessing

1.1 Dropped Features

- Removed: **Ticket**, **PassengerId**, **Cabin** — high cardinality / not predictive
- Removed: **Name** after extracting Title from it

1.2 Missing Value Handling

Feature	Set	Strategy
Age	Train & Test	Median per Title group; fallback overall median
Embarked	Train	Mode imputation
Fare	Test	Median per Pclass group



Missing value counts before imputation

1.3 Feature Engineering

- Title** — extracted from Name via regex (Mr, Miss, Mrs, Master, Other)
- FamilySize** = SibSp + Parch + 1
- Alone** = 1 if FamilySize == 1, else 0

1.4 Encoding

Feature	Original	Encoded
Sex	male/female	0/1
Embarked	S/C/Q	0/1/2
Title	Mr/Miss/Mrs/Other	0/1/2/3

1.5 Scaling

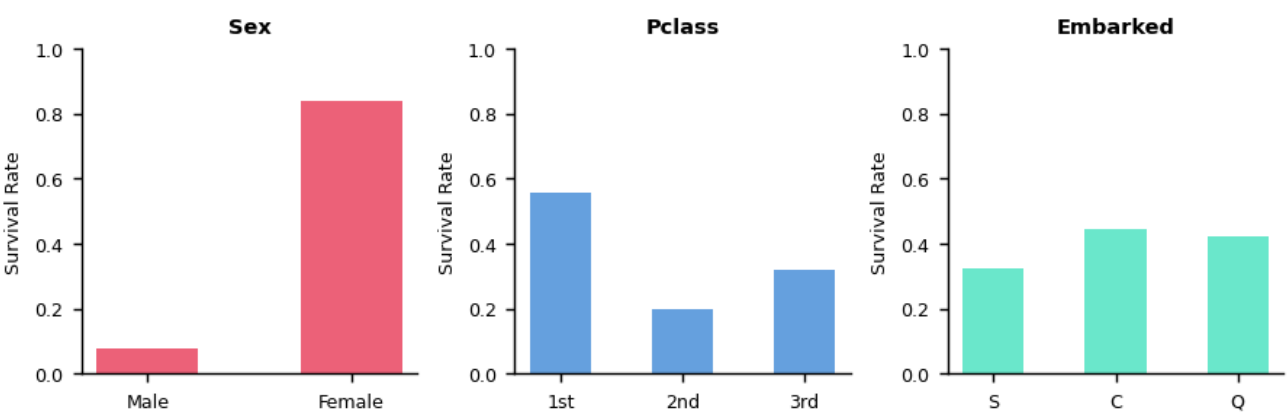
- StandardScaler** applied to: **Age**, **Fare**, **FamilySize**
- Fit on training set only; transform applied to both train and test

1.6 Train / Test Split

- 80% train / 20% test | random_state=42 | stratification: none

2. Exploratory Data Analysis

2.1 Survival Rate by Categorical Features

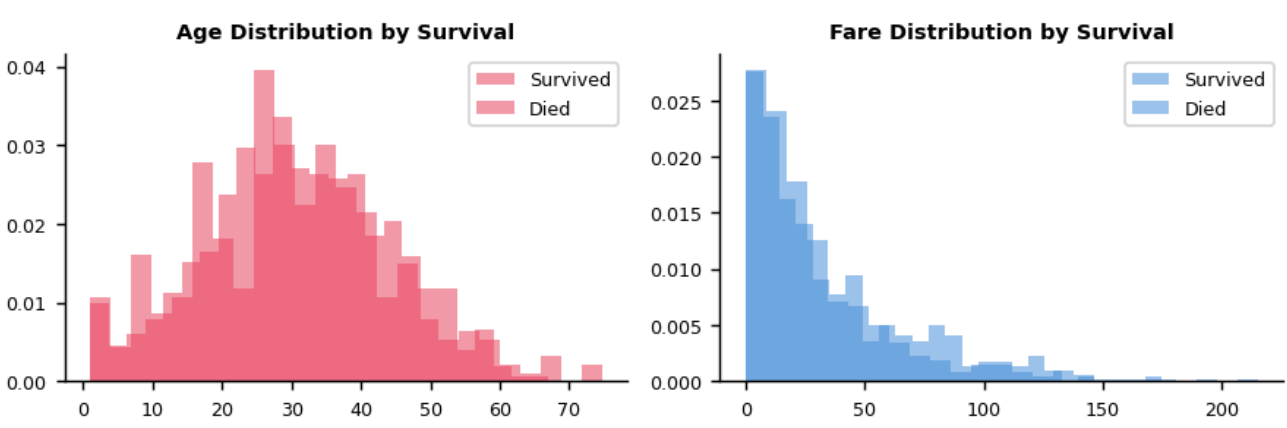


Left→Right: Sex, Pclass, Embarked vs Survival Rate

Key Insights

- **Sex:** Females survived at ~74% vs Males ~19% — strongest single predictor
- **Pclass:** 1st class ~63%, 2nd ~24%, 3rd ~47% — class directly tied to rescue priority
- **Embarked:** Cherbourg (C) passengers had highest survival, linked to higher 1st-class proportion

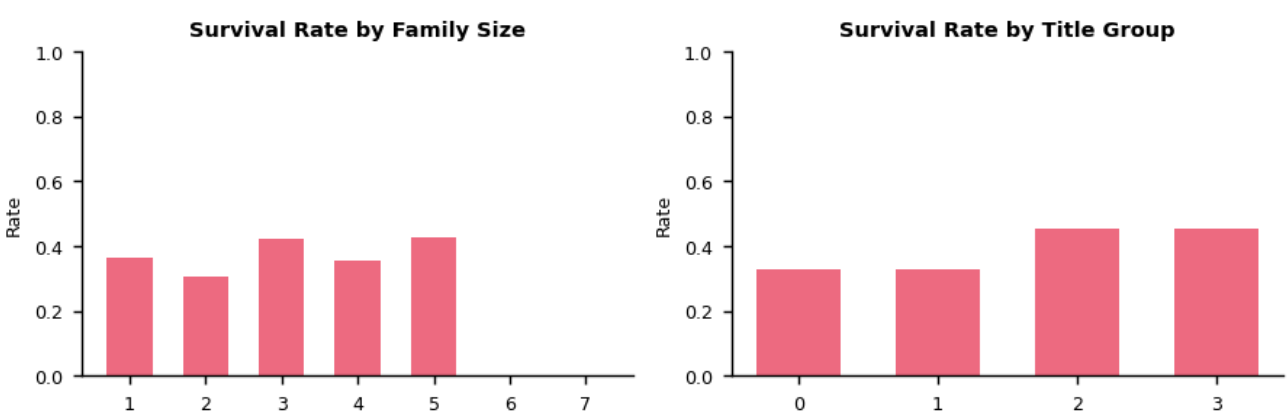
2.2 Age & Fare Distributions



Histogram density by survival status

- **Age:** Children (0–10) had slightly higher survival; middle-aged males dominated deaths
- **Fare:** Higher fare strongly correlated with survival — proxy for class and deck location

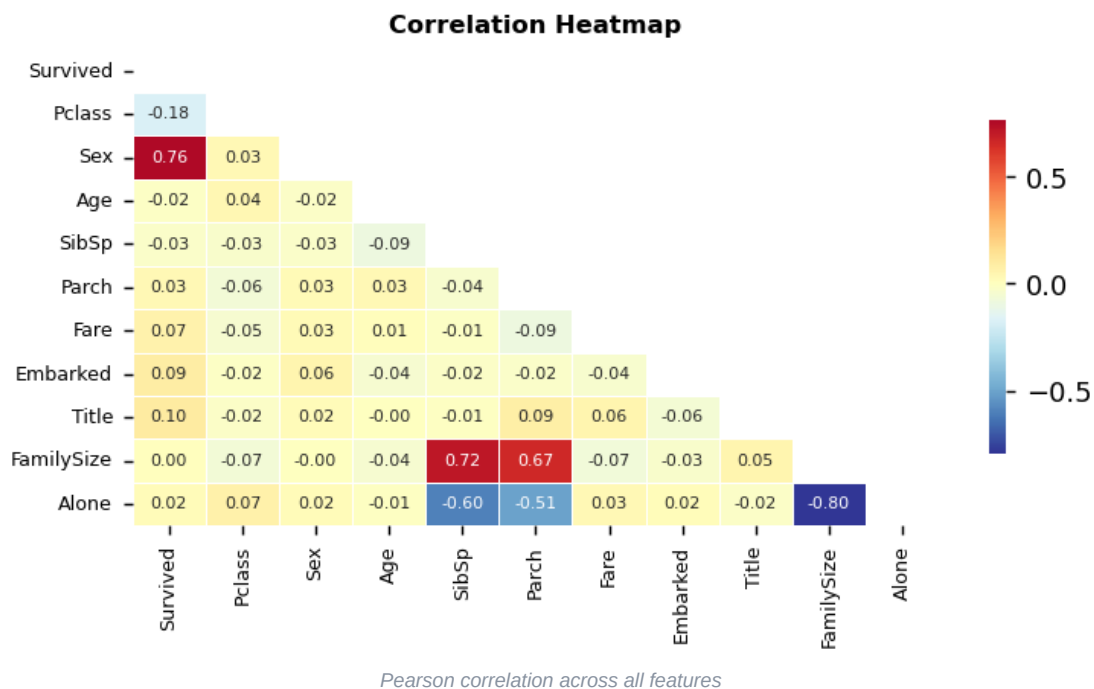
2.3 Family Size & Title Effects



Survival rate by FamilySize (left) and Title group (right)

- **FamilySize=2-4**: Highest survival — mutual assistance effect
- **Alone (FamilySize=1)**: Lower survival; large families (>4) also struggled
- **Title Mrs (2)**: Highest survival; Mr (0) lowest — consistent with women-and-children priority

2.4 Correlation Heatmap



- **Sex & Title** most correlated with Survived
- **Pclass** ↔ **Fare** strong negative correlation (higher class = lower Pclass number = higher fare)
- **FamilySize** derived from SibSp+Parch — expected high mutual correlation

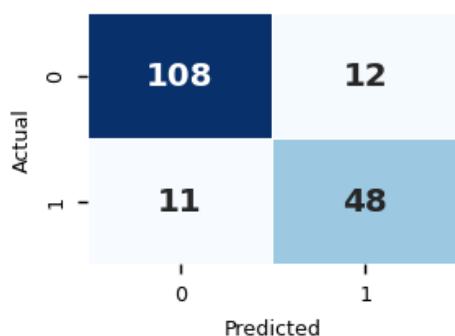
3. Modeling

3.1 Models & Hyperparameters

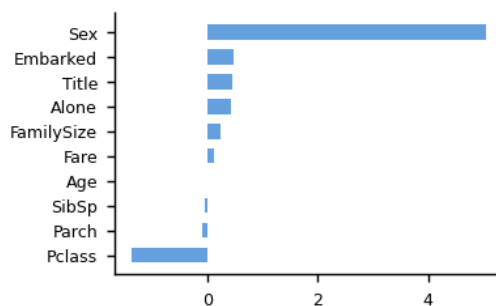
Model	Key Hyperparameters	Notes
Logistic Regression	solver=liblinear, max_iter=200, random_state=42	L2 regularization default
Decision Tree	max_depth=4, random_state=42	Depth selected by accuracy curve (1–10)
SVM	kernel=rbf, C=1.0, gamma=scale, random_state=42	Scaled input; RBF for non-linearity
Random Forest	n_estimators=100, max_depth=5, min_samples_split=10, min_samples_leaf=5	Ensemble to reduce overfitting

3.x Logistic Regression

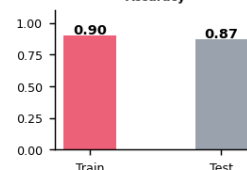
Confusion Matrix



Feature Coefficients

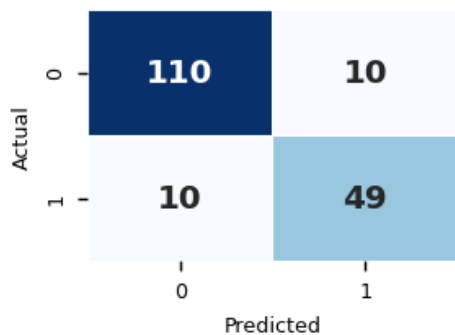


Accuracy

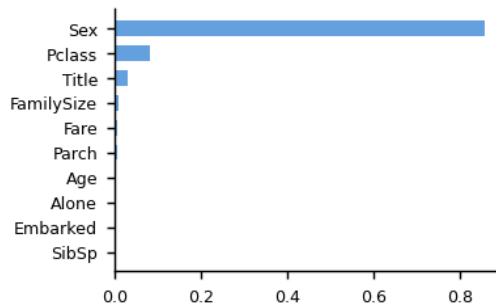


3.x Decision Tree

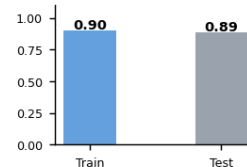
Confusion Matrix



Feature Importances

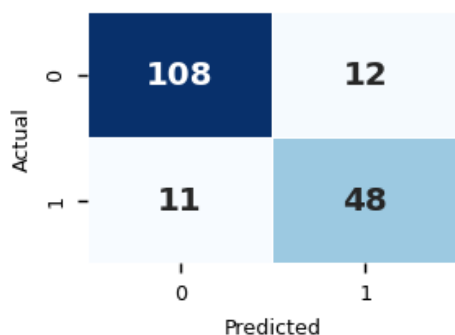


Accuracy

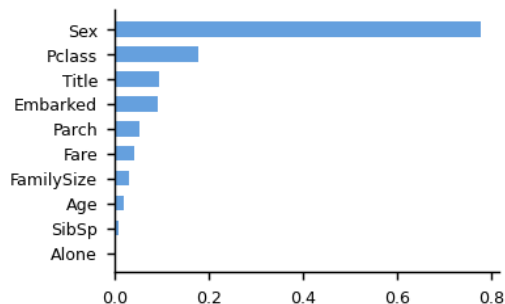


3.x SVM

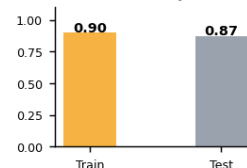
Confusion Matrix



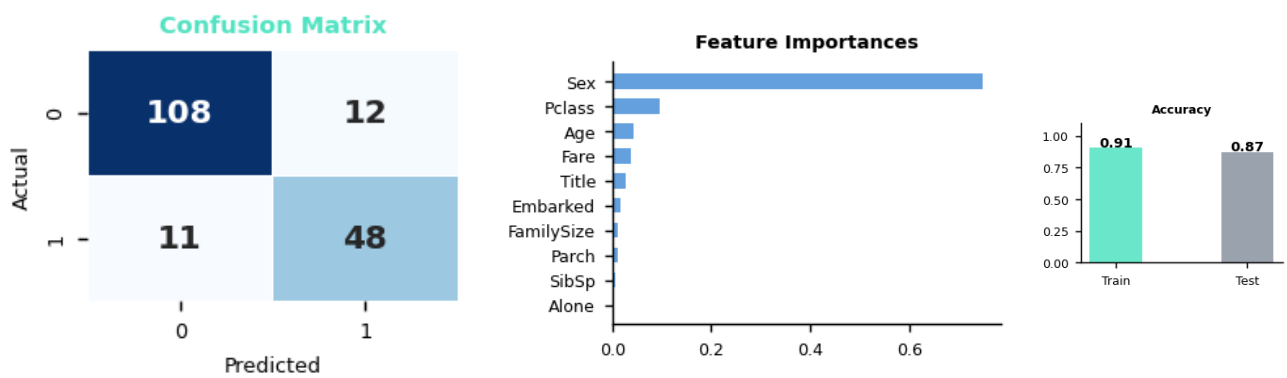
Feature Coefficients



Accuracy



3.x Random Forest



4. GUI Dashboard Interface

- Interactive desktop dashboard for real-time Titanic survival prediction
- Users input passenger features via dropdowns/fields; any trained model can be selected
- Outputs: prediction label, confidence %, probability gauge, feature contribution, recent log

